30401 Mathematics and Statistics - Module 2 (Statistics) - BEMACS

FIRST PARTIAL EXAM - 60 minutes

Surname		Name		Student Number	

I hereby confirm my attendance at the exam.

I declare I have read the Exam Rules and I commit to respect them.

Signature:

Exercise 1 (8 points, R Dataset)

For the following questions, refer to the dataset solvency, contained in the file solvency.rda. A rating agency assigned a score based on the level of long-term solvency to companies registered to the official company registry. A random sample of 90 companies is drawn.

- **ID**: the numerical identifier of the company (one number for each company included in the sample, 1 to 90)
- **score**: score assigned by the rating agency (from 0=minimum level of solvency to 100=maximum level of solvency)
- **sector**: sector in which the company operates (levels: "Financial", "Industrial", and "Other")
- **leverage**: debt to equity ratio, expressed in percentage points
- **age**: number of years the company has been in business
- **listing**: variable indicating whether the company is listed on the stock exchange (Yes) or not (No).

a) (1pt*) Produce and report a table with the possible values taken by the variable **sector** and the relative frequency of each value.

b) (1pt*) What is the median **score** of the companies in the dataset?

- ☐ 57.27
- ☑ 58
- ☐ [60,70]
- ☐ It cannot be calculated

c) (1pt*) Report the percentage of listed companies among each of the three operating sectors.

d) (1pt*) Assess the symmetry of the variable **leverage**. Justify your answer.

e) (1pt**) Calculate and report the conditional mean and standard deviation of the variable **score** for each of the subgroups of companies defined by the three operating **sector**.

f) (2pt**) Compare graphically the distributions of the **score** variable across sectors. In particular, compare the three conditional distributions with respect to: symmetry, centrality and dispersion. Which type of graph did you use for the comparison?

g) (1pt**) Comment on the following statement on the basis of statistical evidence from the data:

"younger companies tend to have a lower solvency level"

Clearly state the graphical and quantitative tools you use.

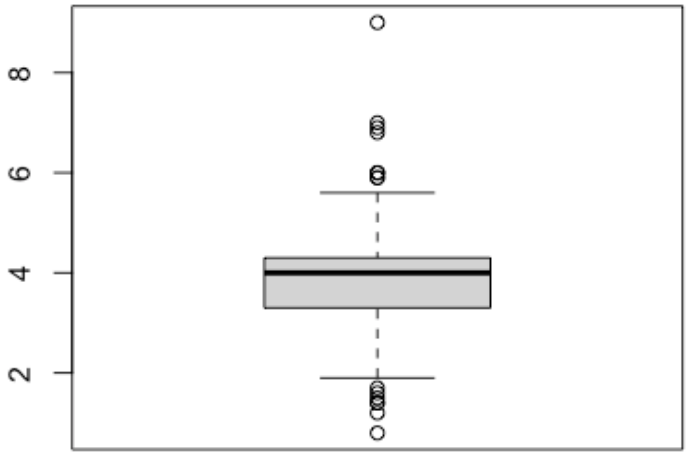
SOLUTION HINT

a)
```
>S=Solvency; tab=table(S$sector); prop.table(tab)
 Financial Industrial      Other
0.2222222  0.4555556  0.3222222
```

c) The required percentages are: 65% for the Financial sector, 31.71% for the Industrial Sector, 20.69% for Other.

```
>prop.table(table(S$sector,S$listing),1)
                    No        Yes
  Financial  0.3500000 0.6500000
  Industrial 0.6829268 0.3170732
  Other      0.7931034 0.2068966
```

d) From the box-plot of leverage below, the variable does not show a clear symmetry.



Digging deeper (not necessary):

The mean (3.903) is lower than the median (4). Indication of left skewness.

The upper part of the box is narrower wrt the lower part. Indication of left skewness.

The upper whisker is slightly narrower than the lower whisker. Indication of left skewness or symmetry.

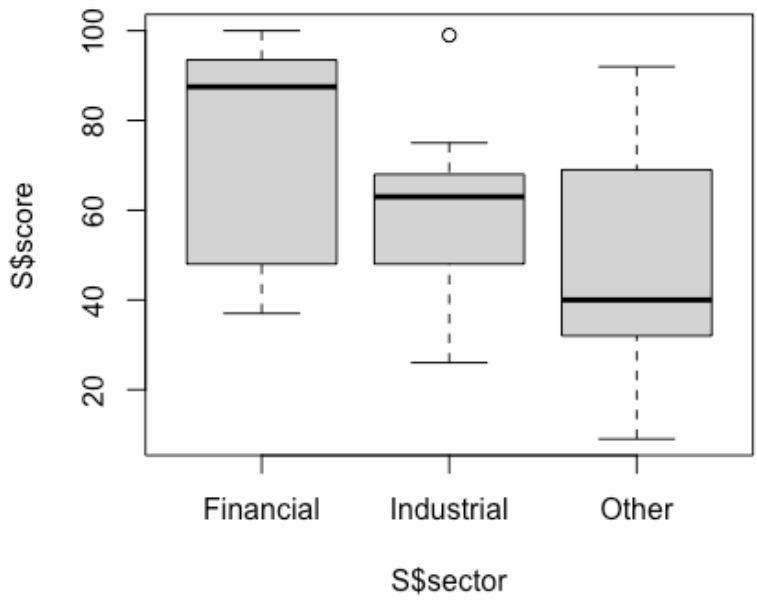
There are similar numbers of upper outliers (9) and lower outliers (7). Dispersion is similar apart from one very high upper outlier. Indication of symmetry.

Hence: the symmetry is not very clearly defined. It could be considered symmetric or slightly left-skewed.

e)

```
> tapply(S$score,S$sector,mean)
 Financial Industrial      Other
  75.00000   55.46341   47.58621
> tapply(S$score,S$sector,sd)
 Financial Industrial      Other
  22.01435   18.24020   26.13361
```

f) For the comparison we can use multiple box plots reporting the conditional distributions of scores for the three different sectors.



Symmetry. The conditional distribution of score seems left skewed for Financial and Industrial, and right skewed for Other.

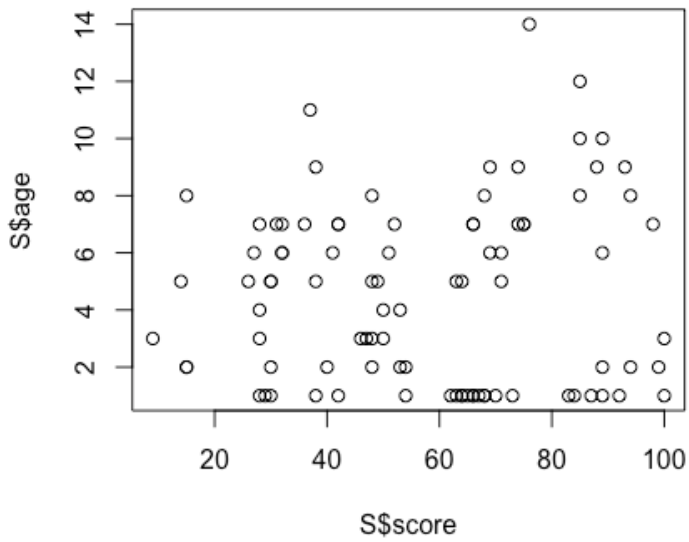
Centrality. Centrality measures (conditional mean as calculated in point e) and median from box plots) are higher for Financial, lower for Industrial and even lower for Other.

Dispersion. We can measure dispersion with conditional standard deviations as calculated in point e, and looking at core and whiskers of box plots. All measure indicate a higher conditional dispersion for the sector Other, similar but lower dispersion for Financial, lower dispersion for Industrial.

g) We can investigate the claim graphically with a scatterplot or by calculating the correlation between score and age.

> cor(S$score,S$age); > plot(S$score,S$age)

[1] 0.05875666



Correlation is close to zero and the scatterplot does not show a strong pattern. There is very little (if any) evidence from the data that can support the claim.

EXERCISE 2 (7 points)

Consider the setting of Exercise 1. For a random sample of 150 companies (**different from the sample available in Solvency.rda**) each company is assigned with a **Solvency Level** from 1 to 6 based on the rating agency's score (from 0 to 100).
Companies with a score between 0 and 30 are assigned with a Solvency Level of 1.
Companies with a score greater than 30 but lower equal than 60 are assigned with a Solvency Level equal to 2,…
In the table below you can find the full classification and the absolute frequency of each solvency level. Raw data is not available for this sample.

Score Class	[0,30]	(30,60]	(60,70]	(70,80]	(80,90]	(90,100]
Solvency Level	**1**	**2**	**3**	**4**	**5**	**6**
Number of Companies	27	48	30	15	18	12

a) (1pt*) Which type of data is Solvency Level?

- ☐ Binary Data
- ☐ Categorical Nominal Data
- ☐ Numerical Count Data
- ☑ Categorical Ordinal Data

b) (1pt**) Compute and report the five number summary (minimum, first quartile, median, third quartile, maximum) of the variable Solvency Level.

c) (1pt*) What is your assessment of the sample distribution of Solvency Level?

- ☐ It is left skewed
- ☑ It is right skewed
- ☐ It is symmetric
- ☐ No clear symmetry assessment is possible

d) (1pt*) What is the percentage of companies with Solvency Level equal to 2 or lower?

☐ 17.8% ☐ 32.2% ☑ 50% ☐ It cannot be calculated

e) (1pt**) Estimate the percentage of companies out of the 150 companies in the sample with a **score** lower than 45. Why is it an estimate and not an exact value?

(R Dataset)

f) (1pt**) **<u>Now consider the sample of 90 companies in Solvency.rda.</u>** Report the number of companies for each Solvency Level, as defined by the score classes in the table above.

g) (1pt**) What is the exact percentage of companies out of the 90 companies with a **score** lower than 45?

SOLUTION HINT

b) Min=1, Q_1=2, Median=2.5, Q_3=4, Max=6

e) There are 48 companies with score in (30,60]. If we assume companies are uniformly distributed in this interval, 24 companies will have score in (30,45] and therefore 24+27 companies (34% of the total) will have a score lower than 45. It is **not** the exact level as we assumed uniform distribution within classes. We can calculate the exact level with raw data.

f) For our dataset.

Score Class	[0,30]	(30,60]	(60,70]	(70,80]	(80,90]	(90,100]
Solvency Level	**1**	**2**	**3**	**4**	**5**	**6**
Number of Companies	16	29	18	8	11	8

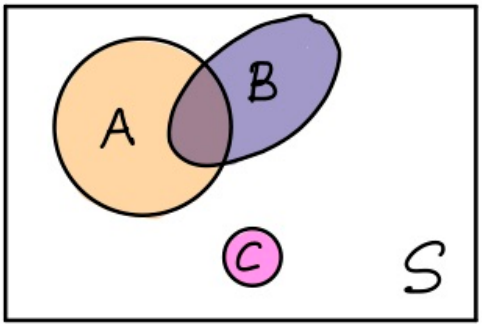
```
> h=hist(S$score,breaks=c(0,30,60,70,80,90,100))
> h$counts
[1] 16 29 18  8 11  8
```

g) The exact percentage is 33.33%.

```
> sum(S$score<45)
[1] 30
```

EXERCISE 3 (6 points)

a) (3pt*) Let A, B and C be 3 events as represented in the Venn diagram below.



For each of the following statements decide whether it is TRUE or FALSE in general.

- $A^c \cap B^c = C$ [TRUE] **[FALSE]**
- $(A^c \cap B)^c = A^c \cup B$ [TRUE] **[FALSE]**
- $(C \cap B^c)^c \supseteq A$ **[TRUE]** [FALSE]
- $P(C^c) \geq P(C)$ [TRUE] **[FALSE]**
- $P(C \cup A \cup B^c) = 1$ [TRUE] **[FALSE]**
- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B)$ **[TRUE]** [FALSE]

b) (3pt*) Starting from the axioms of probability, prove carefully by induction the following result.

Given a probability measure P and the events $A_1, A_2, \ldots, A_n$ such that $A_i \cap A_j = \emptyset$ for $i \neq j$, then $P(\bigcup_{k=1}^{n} A_k) = \sum_{k=1}^{n} P(A_k)$.

EXERCISE 4 (10 points)

In a single request, a user of a web LLM wrapper (think of something like ChatGPT) submits a random number of tokens X that can be modelled according to a Poisson distribution with mean parameter $\mu = 15$ (hundreds tokens), i.e. $X \sim Poisson(\mu = 15)$.

a) (1pt*) What is the probability for a user to submit a request of 20 hundred tokens or more?

☐ 0% ☑ 12.5% ☐ 8.3% ☐ 5.3%

b) (1pt*) 99% of the requests should consist of how many tokens at most?

The cost to respond to a request with x token depends on the number of tokens and it is given by:

$$cost = a + \log(x),$$

where $a = 1$. The cost is expressed in cents.

c) (2pt**) What is the probability that the cost of a request strictly exceeds 3.5 cents?

d) (2pt**) What's the cost of the top 5% most expensive requests?

e) (2pt**) Now assume that a request exceeding (strictly more than) 25 hundreds tokens requires a more costly model, and that the web LLM wrapper receives 10,000 requests in a day. Calculate the probability that during the day only 80 of the requests require the more expensive model at most.

f) (2pt**) Calculate (possibly through simulation, **report the code you use**) the (approximated) expected cost for fulfilling **one** request if:

- The cost of a request with under 25 hundreds tokens is given by $cost = a + \log(x)$, where x is the number of tokens and $a = 1$
- The cost of a request with 25 hundreds tokens or more is given by $cost = b + \log(x)$, where x is the number of tokens and $b = 50$

SOLUTION HINT

b) 99% of the requests should be of 25 hundred tokens or less.

```
> qpois(0.99,15)
[1] 25
```

c) $P(cost > 3.5) = P(1 + log(X) > 3.5) = P(X > \exp(2.5))$

```
> 1-ppois(exp(2.5),15)
[1] 0.732389
```

d) $P(cost \leq q) = 0.95$ if and only if $P(X \leq \exp(q-1)) = 0.95$. Calculate the quantile of order 0.95 of X

```
> qpois(0.95,15)
```

[1] 22

And set $\exp(q-1) = 22$ to get $q = log(22) + 1 = 4.091042$. The cost of the 5% most expensive requests is 4.09 cents.

e) The probability for a request to exceed 25 hundred tokens is $p = 0.006184904$

```
> p = 1-ppois(25,15); p
[1] 0.006184904
```

Model the number of requests exceeding such threshold as a Poisson distribution with mean $p * n$ where $n = 10{,}000$ to obtain that the required probability is 98.9%.

```
> n = 10000
> ppois(80,n*p)
[1] 0.9888589
```

f) Through simulation the expected cost can be approximated as 4.2132.

```
> N = 1000000
> x = rpois(N,15)
> cost = (x<25)*(1+log(x))+(x>=25)*(50+log(x))
> mean(cost)
[1] 4.213253
```